Juan Zurita

Teaching Statement

juan.zurita@ed.ac.uk · juanignaciozurita.github.io/personalwebsite

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I teach economics as a quantitative, computational discipline. My courses—from large introductory lectures to specialist Honours seminars—share a common design principle: students learn economic ideas best when they can build, estimate, and break the models themselves. Over fifteen years at four universities across three countries (Argentina, Australia, and the United Kingdom), I have taught macroeconomics, econometrics, forecasting, and programming to students ranging from first-year undergraduates to PhD candidates. At Edinburgh, I have designed two new courses—Programming and Numerical Methods and Deep Learning for Macroeconomics—that equip economics students with the computational tools their research and careers increasingly demand.

Teaching Philosophy

Three principles guide how I design and deliver a course.

Theory and computation together, not in sequence. Economics students too often encounter mathematical models in lectures and data work in separate lab sessions, with little connection between the two. I structure my courses so that computation is not an add-on but the medium through which students engage with theory. In Programming and Numerical Methods, for instance, students implement a Cobb-Douglas production function, compute its marginal products numerically and via automatic differentiation (JAX), and compare both to the analytical solution—all in a single session. The exercise teaches Python syntax, but more importantly it teaches diminishing returns as something you can see and test, not just prove.

Concrete before abstract. I begin with a specific economic question or dataset before introducing the general framework. In Economics 2 at Edinburgh, I open the fiscal-policy topic not with the IS-LM derivation but with the question "Did the UK furlough scheme work?"—a question students lived through. The formal model enters as the tool that lets them reason about the answer. This sequencing sustains attention and gives students a mental anchor for the algebra that follows.

Active problem-solving in real time. Lectures that consist only of slides watched passively produce shallow retention. I interrupt exposition regularly with short exercises—"predict the output of this code," "which parameter drives this result?"—that force students to commit to an answer before I reveal one. In larger classes I use live QR-code polls to surface misconceptions immediately. The goal is to make confusion visible early, when it is cheapest to fix.

Courses Designed

Programming and Numerical Methods for Economics (Honours, Edinburgh, 2024–present). This course teaches economics students to solve, simulate, and estimate the models they encounter in their Honours dissertations. The syllabus covers Python fundamentals, data manipulation with Pandas, numerical optimisation, root-finding, interpolation, numerical differentiation (including automatic differentiation with

JAX), and Monte Carlo methods—each motivated by an economic application. Students interpolate value-function iterations for a consumption-savings problem, estimate impulse responses from a VAR, and compute numerical elasticities of production functions. All materials—lecture slides, interactive Jupyter notebooks, and problem sets—are open-access on my course website.

Deep Learning for Macroeconomics (Edinburgh). A seminar-style course introducing neural-network methods for solving and estimating macroeconomic models. Students work through interactive notebooks covering feedforward networks, recurrent architectures, and applications to dynamic programming and likelihood-free inference. The course responds to growing demand in the profession for researchers who can bridge machine learning and structural macroeconomics.

Core Teaching Experience

Macroeconomics (undergraduate and postgraduate). I have taught macroeconomics at every level. At Edinburgh, I deliver Economics 2 (the core second-year macro course, 2024–2026) covering national accounts, the IS-LM model, monetary and fiscal policy, and open-economy extensions. At UTS, I taught Intermediate Macroeconomics, Advanced Macroeconomics (Honours), and Macroeconomics 2 for PhD students, where the syllabus covered dynamic programming, incomplete markets, and computational solution methods. At the Catholic University of Argentina, I taught History of Economic Thought (2011–2015), tracing the intellectual development of macroeconomic ideas from the classicals through Keynes to modern DSGE models.

Econometrics and forecasting. At the Australian National University, I taught Economic and Business Forecasting (2023), covering time-series methods, VAR estimation, and forecast evaluation. At Edinburgh, I teach Applications of Econometrics (Honours, 2026), where students apply regression, instrumental-variables, and panel-data methods to replicate published empirical results.

Central banking. At UTS, I taught Central Banking and Monetary Policy (2024), a course that drew directly on my research in monetary-policy transmission and bank balance sheets. Students analysed real central-bank communications, estimated Taylor rules, and debated the distributional effects of quantitative easing.

Supervision

I supervise Honours dissertations at Edinburgh, guiding students through topic selection, data collection, empirical estimation, and interpretation. My supervisees have worked on topics including the macroeconomic effects of housing-market shocks, the pass-through of monetary policy to mortgage rates, and forecasting inflation with machine-learning methods. I find supervision among the most rewarding parts of the job—it is where students transition from learning established techniques to producing original analysis.

Teaching Across Cultures

Teaching in Argentina, Australia, and Scotland has given me experience with very different student populations and academic traditions. At the Catholic University of Argentina, I taught in Spanish to students for whom economics was embedded in broader social-science training. At UTS, many students were international, combining economics with business or IT degrees, and valued applied, industry-relevant skills. At Edinburgh, I teach a mix of Scottish, English, EU, and international students in a research-intensive environment where rigorous analytical training is prized. Adapting to these contexts has taught me that clarity, structure, and genuine interest in student progress translate across any classroom—but that the examples, pacing, and assumed background must be calibrated to the cohort, not imported from the last institution.

What I Would Contribute

I can contribute to a department's teaching mission in three ways. First, I offer reliable coverage of core macro and econometrics at all levels—courses every department needs taught well. Second, I bring specialist courses in computational methods and machine learning for economics that few departments currently offer but many students increasingly need. Third, my research in credit markets and banking generates a natural pipeline of Honours and Master's dissertation topics at the intersection of macro, finance, and applied econometrics.

Juan Zurita
Lecturer in Economics / Early Career Researcher
School of Economics, University of Edinburgh
30 Buccleuch Place, Newington, Edinburgh EH8 9JT, United Kingdom